A Survey on Vision–Language–Action Models for Embodied AI

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***Abstract*—Embodied AI is widely recognized as a cornerstone of artificial general intelligence (AGI) because it involves controlling embodied agents to perform tasks in the physical world. Building on the success of large language models (LLMs) and vision–language models (VLMs), a new category of multimodal models—referred to as vision–language–action (VLA) models—has emerged to address language-conditioned robotic tasks in embodied AI by leveraging their distinct ability to generate actions. The recent proliferation of VLAs necessitates a comprehensive survey to capture the rapidly evolving landscape. To this end, we present the first survey on VLAs for embodied AI. This work provides a detailed taxonomy of VLAs, organized into three major lines of research. The first line focuses on individual components of VLAs. The second line is dedicated to developing VLA-based control policies adept at predicting low-level actions. The third line comprises high-level task planners capable of decomposing long-horizon tasks into a sequence of subtasks, thereby guiding VLAs to follow more general user instructions. Furthermore, we provide an extensive summary of relevant resources, including datasets, simulators, and benchmarks. Finally, we discuss the challenges facing VLAs and outline promising future directions in embodied AI. A curated repository associated with this survey is available at: https://github.com/yueen-ma/Awesome-VLA**

***Index Terms*—Embodied AI, large language model (LLM), multimodality, robotics, vision–language–action (VLA) model, world model.**

I. Introduction

VISION–LANGUAGE–ACTION (VLA) models are a class of multimodal models within the field of embodied AI, designed to process information from three modalities: vision, language, and action. Unlike conversational AI, embodied AI requires controlling physical embodiments that interact with the environment, and robotics is the most prominent domain of embodied AI. In language-conditioned robotic tasks, the policy must possess the capability to understand language instructions, visually perceive the environment, and generate appropriate actions, necessitating the multimodal capabilities of VLAs. The term was recently coined by RT-2 [1]. Compared with earlier deep reinforcement learning (RL) approaches, VLAs offer superior versatility, dexterity, and generalizability in complex environments. As a result, they are well-suited not only for controlled settings like factories but also for everyday tasks in household environments.

Early developments in deep learning primarily consisted of unimodal models. In computer vision (CV), models like AlexNet [2] showcased the potential of artificial neural networks (ANNs). Recurrent neural networks laid the groundwork for numerous natural language processing (NLP) models, but NLP has seen a transition in recent years, with Transformers [3] taking precedence. The deep Q-network (DQN) [4] demonstrated that ANNs can successfully tackle RL problems. Leveraging advancements of unimodal models across diverse machine learning fields, multimodal models now address a wide range of tasks, such as visual question answering (VQA), image captioning, and text-to-video generation.

Conventional robot policies based on RL have largely focused on addressing a limited set of tasks within controlled environments, such as item grasping [5]. However, there is a growing need for more versatile multitask policies, akin to recent trends in large language models (LLMs) and vision–language models (VLMs). Developing a multitask policy is challenging, as it requires learning a broader set of skills and adapting to diverse environments. Furthermore, specifying tasks via language instructions offers a more intuitive user–robot interface, which necessitates the development of language-conditioned robot policies.

Built upon the success of large VLMs, VLA models have demonstrated their potential in addressing these challenges, as illustrated in Fig. 1. Similar to VLMs, VLAs utilize vision foundation models as vision encoders to obtain pretrained visual representations (PVRs) of the current environmental state, such as object class, pose, and geometry. VLAs encode instructions using the token embeddings of their LLMs and employ various strategies to align vision and language embeddings, including approaches like BLIP-2 [6] and LLaVA [7]. By finetuning on robot data, the LLM can function as a decoder to predict actions and perform language-conditioned robotic tasks. These cross-disciplinary innovations drive the advancement of VLAs in embodied AI—a critical building block for AGI.

VLAs are closely related to three lines of work, as depicted by the timelines in Fig. 2(b) and the taxonomy in Fig. 3. Some approaches focus on individual components of VLAs (see Section III-A), such as PVRs, dynamics learning,

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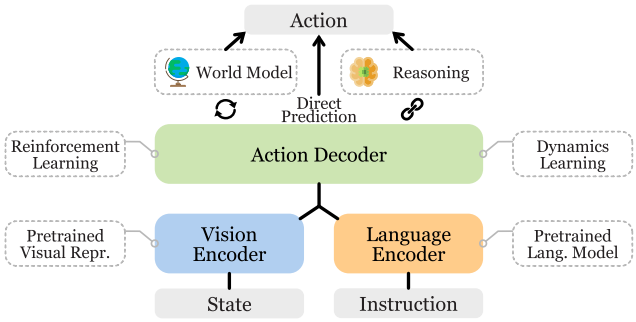

Fig. 1. General architecture of VLA models. Three representative methods for action prediction are shown. Related components are presented in dashed boxes.

world models, and reasoning. Meanwhile, a substantial body of research is dedicated to low-level control policies (see Section III-B). In this category, language instructions and visual observations are fed into the control policy, which then generates low-level actions—such as translation and rotation—thereby rendering VLAs an ideal choice for control policies. In contrast, another category of models serves as high-level task planners responsible for task decomposition (see Section IV). These models break down long-horizon tasks into a sequence of subtasks that, in turn, guide VLAs toward the overall goal, as illustrated in Fig. 4. Most current robotic systems adopt such a hierarchical framework [8], [9], [10] because the high-level task planner can leverage models with high capacity while the low-level control policy can focus on speed and precision, similar to hierarchical RL.

To provide a more comprehensive overview of current progress in embodied AI, we propose a generalized definition of "VLA," as illustrated by the Venn diagram in Fig. 2(a). We define a VLA as any model capable of processing multimodal inputs from vision and language to produce robot actions that accomplish embodied tasks, typically following the architecture in Fig. 1. The original concept of VLA referred to a model that adapts VLMs to robotic tasks [1]. Analogous to the distinction between LLMs and more general language models, we designate the original VLAs as "large VLAs" (LVLAs) because they are based on LLMs or large VLMs.

A. Related Work

To the best of our knowledge, this survey is the first to review the recent progress of VLA models, a rapidly emerging research area. Previous surveys have investigated other facets of embodied AI. Firoozi et al. [11] comprehensively summarized foundation models in robotics up to 2023, while Wang et al. [12] focused on LLMs in robotics. Hu et al. [13] examined more recent vision, language, and robotic foundation models for general-purpose robots. Kawaharazuka et al. [14] concentrated on real-world robot applications. In contrast, our work emphasizes VLA models, thereby complementing and extending the existing literature on embodied AI.

B. Contributions

To the best of our knowledge, this article is the first comprehensive survey of emerging VLA models in the field of embodied AI.

1) *Comprehensive Review:* We present a thorough review of emerging VLA models in embodied AI, covering various aspects including components, architectures, training objectives, and robotic tasks.
2) *Taxonomy:* We introduce a taxonomy of VLAs based on the hierarchical framework of current robotic systems, comprising two hierarchies: a low-level control policy and a high-level task planner. Control policies execute low-level actions based on specified language commands and the perceived environment. Task planners provide guidance to control policies by decomposing long-horizon tasks into subtasks. We also discuss various essential components of VLAs.
3) *Resources:* We summarize the necessary resources for training and evaluating VLAs, including recent datasets and benchmarks in real-world or simulated environments. We also discuss various approaches to address current challenges, such as data scarcity and inconsistency.
4) *Future Directions:* We outline current challenges and promising future opportunities in the field, such as safety, foundation models, and real-world deployment.

II. Background

Embodied AI is a unique form of artificial intelligence that actively interacts with the physical environment. This sets it apart from other AI models, such as conversational AI, which primarily handles textual conversations, or generative AI models that focus on tasks like text-to-video generation. Embodied AI encompasses a broad spectrum of embodiments, including smart appliances, smart glasses, autonomous vehicles, and more. Among them all, robots stand out as one of the most prominent embodiments.

Robot learning is also usually framed as an RL problem, represented as a Markov decision process (MDP) consisting of states (s), actions (a), and rewards (r). An MDP trajectory is denoted by $\tau = (s_1, a_1, r_1, \ldots, s_T, a_T, r_T)$. In certain scenarios, robotic tasks may also be viewed as partially observable Markov decision processes (POMDPs) due to incomplete observations. The primary objective of RL is to train a policy capable of generating optimal actions for the current state $\pi(a_t|s_t)$. Various methods, such as TD learning and policy gradients, can be employed to achieve this. However, in cases where defining the reward function proves challenging, imitation learning is utilized to directly model the action distribution within trajectories devoid of rewards $\tau = (s_1, a_1, \ldots, s_T, a_T)$. Furthermore, many multitask robot models employ language as instructions p to determine which task or skill to execute, leading to the development of language-conditioned robot policies $\pi(a_t|p, s_{\leq t}, a_{<t})$.

III. Vision–Language–Action Models

A. Components of VLA

A body of work focuses on individual components of VLA models, drawing on successes in CV, NLP, and RL. We introduce them through the lens of embodied AI.

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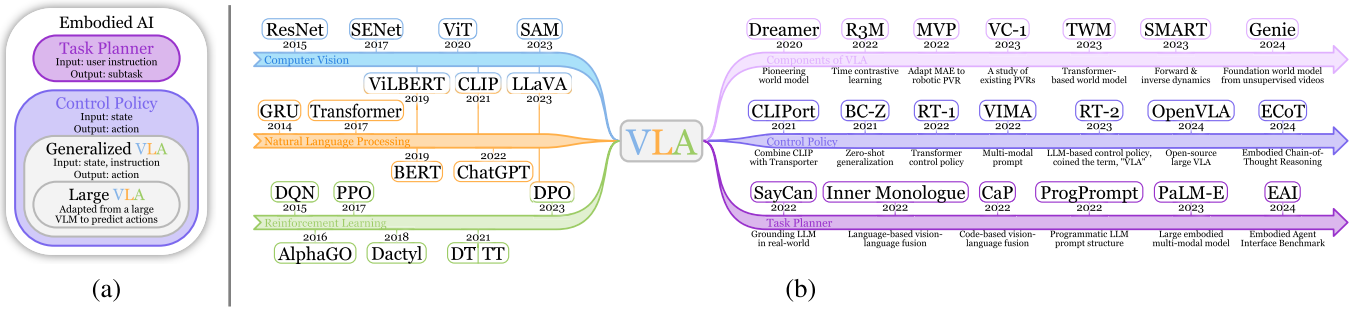

Fig. 2. (a) Venn diagram that outlines the main concepts in embodied AI discussed in this article. (b) Timelines that trace the evolution from unimodal models to VLA models.

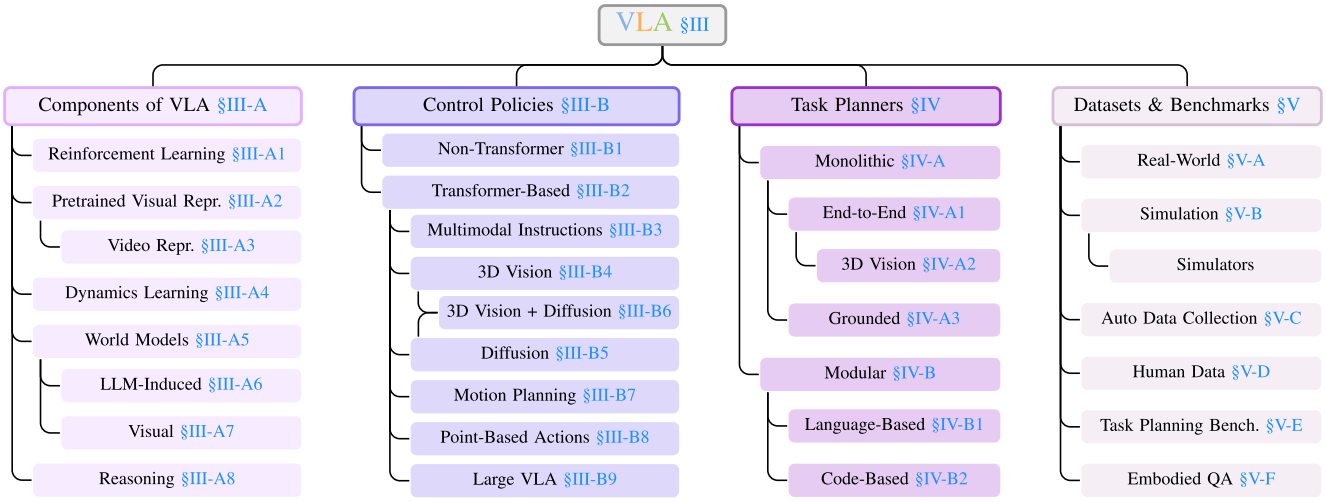

Fig. 3. Taxonomy of VLA models. The organization of this survey follows this taxonomy.

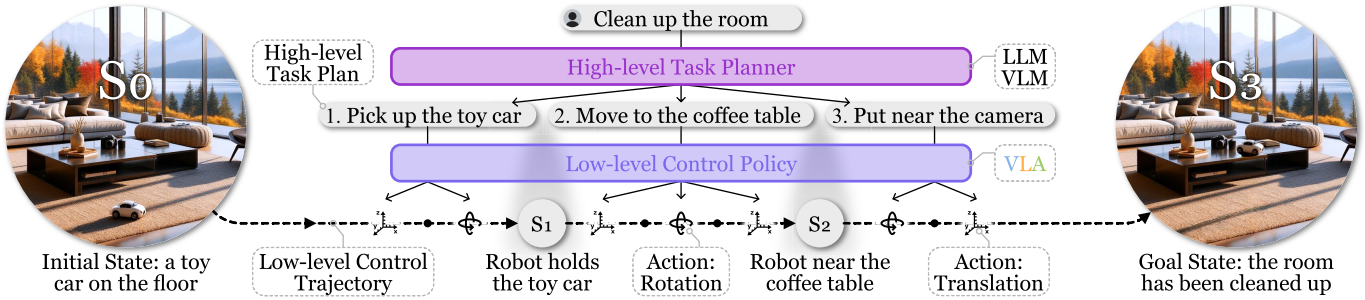

Fig. 4. Illustration of a hierarchical robot policy. The high-level task planner decomposes the user instruction into subtasks, which are then executed step by step by the low-level control policy.

1) Reinforcement Learning: RL laid the foundation for embodied AI and continues to help advance VLAs. DQN first demonstrated learning policies directly from high-dimensional pixel inputs, underscoring the need for greater model capacity in end-to-end RL. RL trajectories—sequences of states, actions, and rewards—naturally align with sequence modeling problems, making them well-suited to Transformer architectures. Pioneering efforts in this direction include decision Transformer (DT) [15] and trajectory Transformer (TT) [16]. Gato [17] further extended this paradigm to a multimodal, multitask, multiembodiment setting. These methods in RL directly inspired modern VLAs [18], [19], which also process multimodal sequences using Transformer-based architectures.

Furthermore, a synergy has emerged between RL and LLMs, benefiting embodied AI in multiple ways. RL from human feedback (RLHF) aligns LLMs with human preferences and has also been applied to robot learning. SEED [20] utilizes RLHF alongside skill-based RL to address the sparse reward issue and to improve safety via human evaluation. Conversely, LLMs also enable novel RL methods. Reflexion [21] proposes a verbal RL framework that replaces weight updates in RL models with linguistic feedback, making it naturally applicable to embodied decision-making where planning also

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TABLE I

PVRS. *V*: VISION. L: LANGUAGE. CL: CONTRASTIVE LEARNING. SIM / REAL: SIMULATED/REAL-WORLD. MANI / NAVI: MANIPULATION/NAVIGATION. FOR SIMPLICITY, WE ONLY SHOW THE MAIN PART OF THE OBJECTIVE EQUATION, OMITTING ELEMENTS SUCH AS TEMPERATURE AND AUXILIARY LOSS. $\mathcal{S}(\cdot)$ DENOTES A SIMILARITY MEASURE

Model	Network	Type	Objective	Notations	Robotic Tasks
CLIP [23]	ViT-B	VL-CL	$\sum_{i=1}^{N} -\log \frac{\exp(\mathcal{S}(x_i,y_i))}{\sum_{j=1}^{N}\exp(\mathcal{S}(x_i,y_j))}$	(x_i, y_i): image-text pair	(Used by CLIPort [29], EmbCLIP [30], and CoW [31])
R3M [24]	ResNet-50	Time-CL	$\sum_{i,j,k} -\log \frac{\exp(\mathcal{S}(x_i,x_j))}{\exp(\mathcal{S}(x_i,x_j))+\exp(\mathcal{S}(x_i,x_k))}$	x_i: video frame at time i; $i<j<k$	**Sim-Mani**: Meta-World, Franka Kitchen, Adroit
MVP [26]	ViT-B/L	MAE	$\sum_{i\in\mathcal{M}} \text{MSE}\left(x_i, f_{\text{mae}}(x_{\notin\mathcal{M}})\right)$	x_i: image patch; $\mathcal{M}$: masked set	**Real-Mani** (xArm7): pick, reach block, push cube, close fridge
VIP [25]	ResNet-50	Time-CL	$\sum_{\tau} -\log \frac{\exp(\mathcal{S}(x_0,x_T))}{\sum_{i,j}\exp(\mathcal{S}(x_i,x_T)-\gamma\mathcal{S}(x_j,x_T))}$	x_i: video frame at time i; $0\le i<j<T$; γ: discount factor; τ: trajectory	**Real-Mani** (Franka): pick and place, close drawer, push bottle, fold towel
VC-1 [32]	ViT-L	MAE, CL	(A combination of previous works, including CLIP, R3M, MVP, and VIP.)	—	**Sim-Mani**: Meta-World, Adroit, DMC, TriFinger, Habitat 2.0; **Sim-Navi**: Habitat
Voltron [33]	ViT-S/B	MAE, Lang-Gen	$\sum_{i\in\mathcal{M}} \text{MSE}\left(x_i, f_{\text{mae}}(x_{\notin\mathcal{M}}, y)\right)$; $\sum_{i=1}^{N} -\log P\left(y_i \mid x, y_{<i}\right)$	x_i: image patch; $\mathcal{M}$: masked set; y: language instruction	**Sim-Mani**: Franka Kitchen; **Real-Mani** (Franka): custom study desk
RPT [28]	ViT	MAE	$\sum_{i\in\mathcal{M}} \text{MSE}\left(x_i, f_{\text{mae}}(x_{\notin\mathcal{M}}, y, z, \ldots)\right)$	x, y, z: three distinct modalities; $\mathcal{M}$: masked set	**Real-Mani** (Franka): stack, pick, pick from bin
DINOv2 [34]	ViT	Self-distillation	$\sum_{x}\sum_{x'\neq x} H(P_t(x), P_s(x'))$	x, x': image views; $H()$: cross-entropy; P_t, P_s: teacher, student	(Used by OpenVLA [35], ReKep [36])
I-JEPA [37]	ViT	JEPA	$\sum_{i\in\mathcal{M}} \text{MSE}\left(f'(x_i), g(f(x_{\notin\mathcal{M}}))\right)$	x_i: image block; $\mathcal{M}$: target set; f, f': encoder, EMA encoder; g: predictor	—
Theia [38]	ViT-T/S/B	Distillation	(Distillation of vision foundation models: ViT, CLIP, SAM, DINOv2, Depth-Anything.)	—	**Sim-Mani&Navi**: CortexBench (VC-1); **Real-Mani**: pick, place, open door/drawer

occurs in language form. Eureka [22] shows that LLMs can design reward functions for embodied agents that outperform expert human-engineered ones. As LLMs evolve, this synergy accelerates progress in RL and embodied AI.

2) Pretrained Visual Representations: The effectiveness of the vision encoder directly influences the performance of VLAs, as it provides crucial information regarding the current state s_t, such as object categories, positions, and affordances. Consequently, numerous methods are devoted to improving the quality of PVRs. Their technical details are compared in Table I.

CLIP [23] has gained widespread adoption as a vision encoder in robotic models. The training objective of CLIP is to identify the correct text-image pair among all possible combinations in a given batch. CLIP is pretrained on the WIT dataset, comprising 400 million image–text pairs. This large-scale training allows CLIP to develop a rich understanding of the relationship between visual and textual information.

R3M [24] proposes two main pretraining objectives: time contrastive learning (CL) and video-language alignment. The objective of time contrastive learning is to minimize the distance between temporally proximal video frames while simultaneously increasing the separation between temporally distant ones. This objective aims to create PVRs that capture the temporal relationships within the video sequence. On the other hand, video-language alignment aims to learn whether a video corresponds to a language instruction. This objective enriches the semantic relevance embedded in the PVRs. VIP [25] also capitalizes on video temporal relationships.

MVP [26] applies a masked autoencoder (MAE) from computer vision (CV) to robotic datasets. The MAE involves masking out a portion of input patches to a ViT model and training it to reconstruct the corrupted patches. This approach closely resembles the masked language modeling technique used in BERT [27] and falls within the purview of self-supervised training. RPT [28] is pretrained with a focus not only on reconstructing visual inputs but also on robotic actions and proprioceptive states.

Voltron [33] introduces a pretraining objective by incorporating language conditioning and language generation into the MAE objective. Employing an encoder–decoder Transformer structure, the pretraining alternates between language-conditioned masked image reconstruction and language generation from masked images. This enhances the alignment between language and vision modalities.

VC-1 [32] conducts an in-depth examination of prior PVRs and introduces an enhanced PVR model by systematically exploring optimal ViT configurations across diverse datasets. In addition, they perform a comprehensive comparative analysis of their model against previous methods on various manipulation and navigation datasets, shedding light on the critical factors that contribute to the improvement of PVRs. Another study [39] also compares previous PVRs obtained under supervised learning or self-supervised learning.

DINOv2 [34] proposes a new self-supervised training paradigm for PVRs that achieves performance beyond that of MAE. It employs a self-distillation framework in which the teacher and student networks receive different views of the same image and match their encoded representations. The student network is updated via gradient descent, while the teacher network is maintained as the exponential moving average of the student network.

I-JEPA [37] is motivated by the joint-embedding predictive architectures proposed by LeCun [40]. It constructs a "primitive" internal world model by comparing the embeddings of patches. Unlike DINO, which uses cropped images, I-JEPA employs masked patches. Moreover, it differs from MAE because it is a nongenerative approach.

Theia [38] proposes distilling various vision foundation models into a single model. By fusing diverse visual information, such as segmentation, depth, and semantics, it outperforms previous PVRs while requiring less data and a smaller model size.

3) Video Representations: Videos are simple sequences of images and can be represented by concatenating the usual PVRs of each frame. However, their multiview nature enables

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TABLE II
DYNAMICS LEARNING METHODS FOR VLAS. $f(\cdot)$ IS THE DYNAMICS MODEL. FWD AND INV: FORWARD AND INVERSE DYNAMICS

Model	Vision Encoder	Type	Objective	Notations	Robotic Tasks
Vi-PRoM [46]	ResNet	Temporal dynamics	$\sum_{t=1}^{T} \mathrm{CE}(t, f(s'_t))$	s'_t: frame in shuffled seq.; $f(\cdot)$ predicts original frame index	**Sim-Mani**: Meta-World, Franka Kitchen; **Real-Mani**: open & close drawer/door
MaskDP [47]	ViT (MAE)	MAE (implicit fwd & inv dynamics)	$\sum_{t=1}^{T} \mathrm{MSE}\left(x_t, f_{\text{mask}}(\tau')\right)$	$x_t \in \{s_t, a_t\}$; τ': masked sequence	**Sim**: DeepMind Control Suite
MIDAS [48]	ViT, Mask R-CNN	Inverse dynamics	$\sum_{t=1}^{T-1} -\log P\left(a_t \mid f_{\text{inv}}(s_{1:T})\right)$	$s_{1:T}$: full state seq.	**Sim-Mani**: VIMA-Bench
SMART [49]	CNN	Forward & inverse dynamics	$\sum_{t=1}^{T-1}\left(\mathrm{MSE}\left(s_{t+1}, f_{\text{fwd}}(s_t, a_t)\right) + \mathrm{MSE}\left(a_t, f_{\text{inv}}(s_t, s_{t+1})\right)\right) + \sum \mathrm{MSE}\left(a_t, f_{\text{mask}}(\tau')\right)$	s_t, a_t: state, action; τ': masked sequence	**Sim**: DeepMind Control Suite
PACT [50]	ResNet-18, PointNet	Forward dynamics	$\sum_{t=1}^{T-1} \mathrm{MSE}\left(s_{t+1}, f_{\text{fwd}}(s_t, a_t)\right)$	s_t, a_t: state, action	**Sim-Navi**: Habitat; **Real-Navi** (MuSHR vehicle)
VPT [51]	ResNet	Inverse dynamics	$\sum_{t=1}^{T} -\log P\left(a_t \mid f_{\text{inv}}(s_{1:T})\right)$	s_t, a_t: state, action	**Sim**: Minecraft
GR-1 [52]	ViT (MAE)	Forward dynamics	$\sum_{t=1}^{T-1} \mathrm{MSE}\left(s_{t+1}, f_{\text{fwd}}(s_t, a_t)\right)$	s_t, a_t: state, action	**Sim-Mani**: CALVIN

a variety of unique representation techniques beyond those mentioned above, such as time CL and MAE. NeRF can be extracted from videos and contains rich 3-D information for robot learning, as exemplified by F3RM [41], and 3D-LLM [42]. The recent 3-D Gaussian splatting (3D-GS) method has been gaining popularity in CV as well as in embodied AI. The 3-D Gaussians can be deformed through physics-based simulation for generative dynamics, as proposed in PhysGaussian [43], or they can serve as 3-D representations for VLMs, as enabled by UniGS [44]. In addition, many videos contain audio, which can provide important cues for robot policies [45].

4) Dynamics Learning: Dynamics learning encompasses objectives aimed at endowing the model $f(\cdot)$ with an understanding of forward or inverse dynamics. Forward dynamics involves predicting the subsequent state resulting from a given action, whereas inverse dynamics entails determining the action required to transition from a previous state to a known subsequent state:

$$\begin{aligned} &\text{Forward dynamics: } \hat{s}_{t+1} \leftarrow f_{\text{fwd}}\left(s_t, a_t\right) \\ &\text{Inverse dynamics: } \hat{a}_t \leftarrow f_{\text{inv}}\left(s_t, s_{t+1}\right). \end{aligned} \tag{1}$$

Some approaches also frame these objectives as reordering problems for shuffled state sequences. Some forward dynamics methods closely resemble the image or video prediction pretraining used in PVRs. We compare them in Table II.

Vi-PRoM [46] presents three distinct pretraining objectives. The first involves a contrastive self-supervised learning objective designed to distinguish between different videos. The remaining two objectives are centered around supervised learning tasks: temporal dynamics learning, aimed at recovering shuffled video frames, and image classification employing pseudo labels. Through a comprehensive comparison with preceding pretraining methods, Vi-PRoM demonstrates its effectiveness for behavior cloning (BC) and PPO.

MIDAS [48] introduces an inverse dynamics prediction task as part of its pretraining. The objective is to train the model to predict actions from observations, formulated as a motion-following task. This approach enhances the model's understanding of the transition dynamics of the environment.

SMART [49] presents a pretraining scheme encompassing three distinct objectives: forward dynamics prediction, inverse dynamics prediction, and randomly masked hindsight control. The forward dynamics prediction task involves predicting the next latent state, while the inverse dynamics prediction task entails predicting the previous action. In the case of hindsight control, the entire control sequence is provided as input, with some actions masked, and the model is trained to recover these masked actions. The incorporation of the first two dynamics prediction tasks facilitates capturing local and short-term dynamics, while the third task is designed to capture global and long-term temporal dependencies.

MaskDP [47] features the masked decision prediction task, wherein both state and action tokens are masked for reconstruction. This masked modeling task is specifically crafted to equip the model with an understanding of both forward and inverse dynamics. Notably, in contrast to preceding masked modeling approaches like BERT or MAE, MaskDP can be applied directly to downstream tasks in a zero-shot manner.

PACT [50] introduces a pretraining objective aimed at modeling state-action transitions. It receives sequences of states and actions as input, and predicts each state and action token autoregressively. This pretrained model serves as a dynamics model, which can then be finetuned for various downstream tasks such as localization, mapping, and navigation.

VPT [51] proposes a video pretraining method that harnesses unlabeled internet data to pretrain a foundation model for the game of Minecraft. The approach begins by training an inverse dynamics model using a limited amount of labeled data, which is then utilized to label internet videos. Subsequently, this newly auto-labeled data is employed to train the VPT foundation model through BC. This methodology follows semi-supervised imitation learning. As a result of this process, the model demonstrates human-level performance across a multitude of tasks.

GR-1 [52] introduces video prediction pretraining tailored for a GPT-style model. The ability to anticipate future frames aligns with forward dynamics learning, contributing to more accurate action prediction.

5) World Models: A world model $P(\cdot)$ encodes commonsense knowledge about the world and predicts the future state for a given action [40]:

$$\hat{s}_{t+1} \sim P\left(s_{t+1} \middle| s_t, a_t\right). \tag{2}$$

It enables model-based control and planning for embodied agents, as they can search for an optimal action sequence in

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imaginary space before executing any real actions. Although forward dynamics learning also attempts to predict the next state, it is usually treated as a pretraining task or an auxiliary loss to enhance the action decoder for the primary robotics tasks, rather than serving as a standalone module. In addition, new embodied experiences can be sampled from visual world models that explicitly generate images or videos of future states.

Dreamer [53] employs three primary modules to construct a latent dynamics model: a representation model, responsible for encoding images into latent states; a transition model, which captures transitions between latent states; and a reward model, predicting the reward associated with a given state. Under the actor-critic framework, Dreamer utilizes an action model and a value model to learn behavior through imagination by propagating analytic gradients through the learned dynamics. Building upon this foundation, DreamerV2 [54] introduces a discrete latent state space along with an improved objective. DreamerV3 [55] extends its focus to a broader range of domains with fixed hyperparameters. DayDreamer [56] applies this method to physical robots performing real-world tasks.

IRIS [57] employs a GPT-like autoregressive Transformer as the foundation of its world model, with a VQ-VAE serving as the vision encoder. Subsequently, a policy is trained using imagined trajectories, which are unrolled from a real observation by the world model. TWM [58] also investigates the application of Transformers in building world models.

6) LLM-Induced World Models: LLMs encompass a wealth of commonsense knowledge about the world, prompting many approaches to leverage that knowledge for improving VLAs.

DECKARD [59] prompts an LLM to generate abstract world models (AWMs) represented as directed acyclic graphs [60], [61], specifically tailored for the task of item crafting in Minecraft. DECKARD iterates between two phases: in the Dream phase, it samples a subgoal guided by AWM; in the Wake phase, DECKARD executes the subgoal and updates the AWM through interactions with the game. This guided approach enables DECKARD to achieve markedly faster item crafting compared with baselines lacking such guidance.

LLM-DM [62] uses an LLM to construct world models in planning domain definition language (PDDL)—a capability that LLM+P [63] did not achieve, as its PDDL world model was hand-crafted. The LLM also acts as an interface, mediating between the generated PDDL model and corrective feedback from syntactic validators and human experts. Finally, the PDDL world model serves as a symbolic simulator, enabling plan generation.

RAP [64] repurposes an LLM to act as both a policy that predicts actions and a world model that provides the state transition distribution. Unlike previous chain-of-thought (CoT) prompting methods, RAP incorporates Monte Carlo tree search (MCTS) to enable structured planning, allowing the LLM to build a reasoning tree incrementally. This reasoning strategy helps RAP find a high-reward path that balances exploration and exploitation. Tree-Planner [65] improves the efficiency by prompting the LLM only once to generate diverse paths within an action tree.

LLM-MCTS [66] extends the problem setting to POMDPs. As a world model, the LLM generates the initial belief of the current state; as a policy, it serves as a heuristic to guide action selection. By leveraging its commonsense knowledge, the LLM reduces the search space of MCTS, thereby improving search efficiency.

7) Visual World Models: Unlike LLM-induced world models, which are in textual form, visual world models generate images, videos, or 3-D scenes of future states—aligning more closely with the physical world. They can further be utilized to generate new trajectories. This direction has been gaining increasing attention since Sora demonstrated world-simulation capabilities [67], as investigated by a dedicated survey [68].

Genie [69] introduces a new class of generative models, termed generative interactive environments. It consists of three main components: a spatiotemporal video tokenizer, an autoregressive dynamics model, and a latent action model. After being trained on unlabeled videos in an unsupervised manner, Genie allows users to interact with the generative environment on a frame-by-frame basis. Consequently, Genie establishes a foundation world model.

3D-VLA [70] proposes a 3-D world model capable of goal generation. It processes visual inputs, such as images, depth maps, and point clouds, and then generates a goal state—either as an image or a point cloud—using diffusion models in response to the user's query. The generated goal state can subsequently be used to guide robot control.

UniSim [71] builds a generative model based on real-world interaction videos. It is capable of simulating visual outcomes for both high-level and low-level actions, which can then be leveraged as new experiences for training embodied agents. E2WM [72] even treats existing simulators as world models to collect embodied experiences via MCTS.

8) Reasoning: Reasoning has become a key capability of LLMs, as demonstrated by CoT methods [73], [74]. In embodied AI, researchers are exploring how to leverage CoT reasoning to refine the decision-making process.

Reasoning is naturally compatible with high-level task planning, as both often occur in the language domain. ThinkBot [75] applies CoT to recover missing action descriptions in sparse human instructions, thereby enhancing instruction coherence and boosting success rates in difficult tasks. ReAct [76] interleaves reasoning traces and actions, where CoT can help create action plans, inject commonsense knowledge, and handle exceptions, ultimately improving embodied decision-making. RAT [77] integrates CoT with retrieval-augmented generation (RAG) to mitigate hallucinations and thus improve embodied planning. Tree-Planner [65] employs a tree-of-thoughts approach for task planning.

Another paradigm involves equipping low-level control policies with reasoning capabilities, as pioneered by ECoT [78]. This method trains OpenVLA [35] to conduct embodied CoT reasoning about plans, subtasks, motions, and visual features, before predicting low-level actions. By relying on this multistep reasoning rather than the "muscle memory" of VLAs, it improves success rates on challenging generalization tasks without requiring additional robot data. CoT-VLA [79] introduces visual CoT reasoning for VLAs.

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9) Strengths and Limitations:

a) PVRs: Although time CL with videos and text-guided pretraining methods, such as CLIP, provide image-level information, they lack the pixel-level details offered by MAE-based self-supervised learning methods. Pixel-level information contains rich details—including segmentation masks, object positions, and depth estimation—that are generally more useful for robot manipulation tasks requiring high precision, as demonstrated by VC-1's comparison [32]. DINOv2 learns both pixel- and image-level features by combining masked image modeling with a momentum encoder and multicrop augmentation, with benefits on downstream robotic tasks evidenced by OpenVLA [35]. I-JEPA focuses on patches in the representation space and, as a result, captures low-level image features more effectively than view-invariance methods such as DINO. Theia distills various off-the-shelf vision foundation models into a single model that surpasses isolated individual models, as demonstrated by comprehensive evaluations against most existing PVRs in robot learning [38].

b) Forward and Inverse Dynamics: Forward dynamics learning is generally more challenging than inverse dynamics learning because predicting future states is more complex than predicting past actions. Consequently, the difficulty of forward dynamics often leads to greater performance improvements, as demonstrated in SMART. However, inverse dynamics models can be used to generate action labels for datasets that contain only states, such as raw robot manipulation videos.

c) World Models and Reasoning: Although both world models and reasoning methods can be applied to low-level control policies and high-level task planners, current approaches remain distinct. World models are predominantly used to interact with control policies because they excel at generating the immediate next state given low-level actions. In contrast, CoT-based reasoning methods focus on task planning since they express thought chains in text, making them well-suited for refining text-based task plans.

B. Low-Level Control Policies

Through the integration of an action decoder with perception modules, such as vision encoders and language encoders, a VLA model π_θ with parameters θ is formed as a control policy to execute language instructions p:

$$\hat{a}_t \sim \pi_\theta \left(a_t \middle| p, s_{\leq t}, a_{<t} \right). \tag{3}$$

It can also be referred to as a low-level policy, low-level controller, or action primitive. The diversity among VLAs arises from individual modules and overall architectures. The general architecture is shown in Fig. 1. This section explores various approaches to designing low-level control policies. Table III summarizes their technical details.

1) Non-Transformer Control Policies: Prior to the adoption of Transformer models, early control policies for language-conditioned robotic tasks varied significantly in architecture.

CLIPort [29] integrates CLIP with the Transporter network, creating a two-stream architecture. The CLIP vision encoder extracts "semantic" information from the RGB image, while the Transporter network extracts "spatial" information from the RGB-D image. The CLIP sentence encoder encodes the language instruction and guides the output SE(2) action, consisting of paired pick and place end-effector poses. It represents an early demonstration of language-conditioned pick-and-place capabilities.

BC-Z [80] processes two types of task instructions: a language instruction or a human demonstration video. The environment is presented to the model in the form of an RGB image. Then the instruction embedding and the image embedding are combined through the FiLM layer, culminating in the generation of actions. This conditional policy is asserted to exhibit zero-shot task generalization to unseen tasks.

MCIL [81] represents a pioneering robot policy that integrates free-form natural language conditioning. This is in contrast to earlier approaches that typically rely on conditions in the form of task IDs or goal images. MCIL introduces the capability to leverage unlabeled and unstructured demonstration data. This is achieved by training the policy to follow either image or language goals, with a small fraction of the training dataset consisting of paired image and language goals.

HULC [82] introduces several techniques aimed at enhancing robot learning architectures. These include a hierarchical decomposition of robot learning, a multimodal Transformer, and discrete latent plans. The Transformer learns high-level behaviors, hierarchically dividing low-level local policies and the global plan. In addition, HULC incorporates a visuolingual semantic alignment loss based on CL to align VL modalities. HULC++ [83] further integrates a self-supervised affordance model. This model guides HULC to the actionable region specified by a language instruction, enabling it to fulfill tasks within this designated area.

UniPi [84] treats the decision-making problem as a text-conditioned video generation problem. To predict actions, UniPi generates a video based on a given text instruction, and actions are extracted from the frames of the video through inverse dynamics. This innovative policy-as-video formulation offers several advantages, including enhanced generalization across diverse robot tasks and the potential for knowledge transfer from internet videos to real robots.

2) Transformer-Based Control Policies: Since the introduction of Transformers, control policies have converged to Transformer-based architectures.

Interactive language [86] presents a robotic system wherein the low-level control policy can be guided in real-time by human instructions conveyed through language, enabling the completion of long-horizon rearrangement tasks. The efficacy of such language-based guidance is primarily attributed to the utilization of a meticulously collected dataset containing diverse language instructions, which surpasses previous datasets by an order of magnitude in scale.

Hiveformer [87] places significant emphasis on leveraging multiview scene observations and maintaining the full observation history for a language-conditioned policy. This approach represents an advancement over previous systems, such as

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TABLE III

VLAs THAT SERVE AS LOW-LEVEL CONTROL POLICIES. ◊ INDICATES LVLAs. CLOSELY RELATED (NON-VLA MODELS) ARE INCLUDED IN BRACKETS. THE REMAINING ARE GENERALIZED VLAs. CONT/DISC: CONTINUOUS/DISCRETE ACTION. TFM: TRANSFORMER. XATTN: CROSS-ATTENTION. CONCAT: CONCATENATION. QUANT.: QUANTIZATION. *p*/*s*: PROMPT/STATE VISION ENCODER. [SC]: SELF-COLLECT DATA

Model	Vision Encoder	Language Encoder / VLM	Action Decoder/Head	Architecture	Training Objectives	Training Datasets	Environments, Embodiments, Tasks, and Skills
CLIPort [29]	CLIP-ResNet50, Transporter-ResNet	CLIP-GPT	LingUNet	Hadamard	BC ($SE(2)$)	[SC]	**Sim**: Ravens
MCIL [81]	Custom-CNN	USE	RNN	LMP	MCIL	Play-LMP, [SC]	**Sim**: 3D Playroom environment
HULC [82], [83]	MCIL CNN	Sentence-BERT	RNN	LMP	MCIL	CALVIN data	**Sim**: CALVIN
Language costs [85]	CLIP-ViT, UNet	CLIP-GPT	STORM	Concat	MLE (cost map)	[SC]	**Sim** & **Real** (Franka): pick, place
Interactive Language [86]	ResNet	CLIP-GPT	TFM	Xattn	BC (cont)	[SC: Language-Table]	**Sim** & **Real** (xArm6): rearrangement
Hiveformer [87]	UNet	CLIP-GPT	TFM, UNet	Concat	BC (cont)	RLBench data	**Sim**: RLBench
PerAct [88]	3D CNN	CLIP-GPT	PerceiverIO	Concat	3D affordance	RLBench data, [SC]	**Sim**: RLBench; **Real** (Franka): pick, place, stack, open drawer, sweep, insert peg, etc.
Act3D [89]	CLIP-ResNet50	CLIP-GPT	TFM	Xattn	3D affordance (voxel)	RLBench data, [SC]	**Sim**: RLBench; **Real** (Franka): reach target, wipe, stack, etc.
RVT [90], RVT-2 [91]	CLIP-ResNet50	CLIP-GPT	TFM	Concat	2D affordance (project to 3D)	RLBench data, [SC]	**Sim**: RLBench; **Real** (Franka): stack, press, place
RoboPoint [92]	ViT-L/14	Vicuna-V1.5 13B	—	Concat	2D affordance (project to 3D)	[SC]	**Real** (Franka): pick, place
Gato [17]	ViT	Sent.Piece	TFM	Concat	BC (cont & disc)	[SC]	**Sim** & **Real** (Sawyer): RGB-stacking
(RoboCat) [93]	VQ-GAN (p, s)	—	TFM	Quant.	BC, observation prediction	Self-improvement	**Sim** & **Real** (Sawyer, Franka, KUKA): stacking, building, lifting, insertion, removal
VIMA [94]	ViT, Mask R-CNN	T5	TFM	Xattn	BC ($SE(2)$)	[SC:VIMA-Data]	**Sim** (Ravens): VIMA-Bench
BC-Z [80]	ResNet18 (p, s)	USE	MLP	FiLM	BC (cont)	[SC]	**Real** (EDR): pick-place/wipe/drag, grasp, push
RT-1 [95]	EfficientNet	USE	TFM	FiLM	BC (disc)	[SC: Fractal]	**Real** (EDR): pick-place, move, knock
MOO [94]	OWL-ViT (p), EfficientNet (s)	USE	TFM	FiLM	BC (disc)	[SC]	**Real** (EDR): pick, move near, knock, place upright, place into
Q-Transformer [96]	EfficientNet	USE	TFM	FiLM	TD error	Fractal, Auto-collect	**Sim**: pick; **Real** (EDR): pick, place, open/close drawer, move near
(RT-Trajectory) [97]	EfficientNet	—	TFM	—	BC (disc)	[SC]	**Real** (EDR): pick, place, fold towel, swivel chair, etc.
(ACT) [98]	ResNet18	—	CVAE-TFM	—	BC (cont, action chunking)	[SC] with ALOHA	**Sim**: transfer cube, bimanual insertion; **Real** (ViperX, WidowX): slot battery, open cup, etc
RoboAgent / MT-ACT [99]	CNN	—	CVAE-TFM	FiLM	BC (cont, action chunking)	RoboSet	**Real** (Franka): pick, place, open/close drawers, pour, push, drag, etc.
RoboFlamingo [100]	CLIP-ViT-L/14	LLaMA, MPT, GPT-NeoX	LSTM, TFM	Xattn	BC (cont)	CALVIN data	**Sim**: CALVIN
RoboUniView [101]	UVFormer	(Model design follows RoboFlamingo)			BC (cont)	CALVIN data	**Sim**: CALVIN
DeeR-VLA [102]	(Model design follows RoboFlamingo++)				BC (cont)	CALVIN data	**Sim**: CALVIN
Instruct2Act [103]	CLIP, SAM	ChatGPT	Robot APIs	Tool-Use	—	—	**Sim** (Ravens): VIMA-Bench
VoxPoser [104]	ViLD, MDETR, OWL-ViT, SAM	GPT-4	MPC	Tool-Use	—	—	**Sim**: Sapien; **Real** (Franka): move&avoid, set table, close drawer, open bottle, sweep trash
UniPi [84]	Imagen video	T5-XXL	CNN	Xattn	Inverse dynamics	[SC:PF], Ravens, BridgeV1	**Sim**: Painting Factory (PF), Ravens
(Diffusion Policy) [105]	ResNet18	—	UNet, TFM	—	DDPM	[SC]	**Sim**: Robomimic, Franka Kitchen, etc; **Real** (UR5, Franka): push-T, flip mug, pour sauce
(DP3) [106]	DP3 Encoder	(Model design follows Diffusion Policy)			DDIM	[SC]	**Sim**: (72 tasks); **Real** (Franka, Allegro hand): roll-up, dumpling, drill, pour
SUDD [107]	ResNet18	CLIP-GPT	UNet, TFM	Concat	DDPM	Lang-guided data generation	**Sim**: MuJoCo; **Real** (UR5e): pick, place
Octo [108]	CNN	T5-base	TFM	Concat	DDPM	OXE	**Real**: BridgeV2, CMU Baking, Stanford Coffee, Berkeley Peg Insert, etc.
3D Diffuser Actor [109]	(Model design follows Act3D)				DDPM	RLBench, CALVIN, [SC]	**Sim**: RLBench, CALVIN; **Real** (Franka): put, open, close, stack, etc.
MDT [110]	CLIP-ViT-B/16 (p), Voltron/ResNet18 (s)	CLIP-GPT	DiT	Concat	DDIM	CALVIN, LIBERO, [SC]	**Sim**: CALVIN, LIBERO; **Real** (Franka): pick, push, open, close, etc.
RDT-1B [111]	SigLIP	T5-XXL	DiT	Xattn	DDPM	(Aggregated Datasets)	**Real** (ALOHA dual-arm): wash, pour, fold, etc.
RT-2 [1] ◊	ViT-4B, ViT-22B	PaLI-X, PaLM-E	Symbol-tuning	Concat	BC (disc), Co-fine-tuning	Fractal, VQA	**Sim**: Language-Table; **Real**: RT-1 evaluation tasks
RT-H [112] ◊	(Model design follows RT-2)				BC (disc)	Diverse+Kitchen	**Real**: Diverse+Kitchen eval tasks
RT-X [113] ◊	(Models from RT-1 and RT-2)				BC (disc)	[SC: OXE]	**Real**: BridgeV2, RT-1 evaluation tasks, etc.
OpenVLA [35] ◊	DINOv2, SigLIP	Prismatic-7B	Symbol-tuning	Concat	BC (disc)	OXE, DROID	**Real**: BridgeV2, RT-1 evaluation tasks, Franka-Tabletop, DROID, etc.
OpenVLA-OFT [114] ◊	(Improves OpenVLA with OFT recipe)				BC (cont, parallel decode w/ chunk.)	LIBERO, [SC]	**Sim**: LIBERO; **Real** (ALOHA setup): fold, scoop, put
TraceVLA [115] ◊	(Model design follows OpenVLA, adding visual trace prompting)				BC (disc)	BridgeV2, Fractal, [SC]	**Sim**: SimplerEnv; **Real** (WidowX): pick, push, fold, swipe
π_0 [116] ◊	SigLIP	PaliGemma	Action expert	Concat	Flow matching	OXE, [SC: π-cross-embod.]	**Real** (general robot control): sweep, open, pack, etc.
RoboMamba [117] ◊	CLIP-ViT	Mamba	MLP	Concat	BC (cont)	[SC]	**Sim**: Sapien; **Real** (Franka): open door/box, staple
SpatialVLA [118] ◊	SigLIP, Ego3D Position Encoding	PaliGemma 2	MLP	Concat	BC (Adaptive Action Grids)	OXE, BridgeV2, Fractal, RH20T	**Sim**: SimplerEnv, LIBERO; **Real** (WidowX): pick, place, close, push
TinyVLA [119] ◊	ViT	Pythia	Diffusion Policy	Concat	DDPM	OXE, [SC]	**Sim**: MetaWorld; **Real** (Franka, UR5): place, stack, flip mug, close drawer, open box
CogACT [120] ◊	DINOv2, SigLIP	LLaMA 2	DiT	Concat	DDIM	OXE, [SC]	**Sim**: SimplerEnv; **Real** (Realman, Franka): pick, place, stack, open/close oven
DexVLA [18] ◊	ViT, ResNet-50	Qwen2-VL 2B, DistilBERT,	ScaleDP	Concat	DDPM	[SC]	**Real** (Franka, UR5e, AgileX): pick, fold shirt, bus table, pour
HybridVLA [121] ◊	DINOv2, SigLIP, CLIP	LLaMA 2, Phi-2	MLP	Concat	DDIM	OXE, DROID	**Sim**: RLBench; **Real** (Franka, AgileX): pick-place, pour, open drawer, fold, etc.
LAPA [122] ◊	C-ViViT, VQ-GAN	LWM-Chat-1M	MLP	Concat	LAPA	OXE, BridgeV2, Something V2	**Sim**: Language-Table, SimplerEnv; **Real** (Franka): pick, cover with towel, knock over
WorldVLA [123] ◊	VQ-GAN	Chameleon	TFM	Quant.	BC (disc, chunk.), world model	LIBERO, OpenVLA data	**Sim**: LIBERO
UniVLA [124] ◊	MoVQ	Emu3	TFM	Quant.	BC (disc, chunk.), world model	(Sim benchmark data)	**Sim**: CALVIN, LIBERO, SimplerEnv

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CLIPort and BC-Z, that only use the current observation. Notably, Hiveformer stands out as one of the early adopters of Transformer architecture as its policy backbone.

Gato [17] proposes a model that can play Atari games, caption images, and stack blocks, all with a single set of model parameters. This achievement is facilitated by a unified tokenization scheme, harmonizing the input and output across diverse tasks and domains. Consequently, Gato enables the simultaneous training of different tasks. Astra [19] optimizes this architecture via trajectory attention.

RoboCat [93] proposes a self-improvement process designed to enable an agent to rapidly adapt to new tasks with as few as 100 demonstrations. This self-improvement process iteratively finetunes the model and self-generates new data with the finetuned model. Built upon the Gato model, RoboCat incorporates the VQ-GAN image encoder. During training, RoboCat predicts not only the next action but also future observations. The effectiveness of the self-improvement process is demonstrated through comprehensive experiments conducted in both simulated and real-world environments under multitask and multiembodiment settings.

RT-1 [95], developed by the same team as BC-Z, shares similarities with BC-Z but introduces some key distinctions. Notably, RT-1 employs a vision encoder based on the more efficient EfficientNet, departing from BC-Z's use of ResNet. However, RT-1 does not use video as a task instruction. In addition, RT-1 replaces the MLP action decoder in BC-Z with a Transformer decoder, producing discretized actions. This modification enables RT-1 to attend to past images, enhancing its performance compared with BC-Z.

Q-Transformer [96] extends RT-1 by introducing autoregressive Q-functions. In contrast to RT-1, which learns expert trajectories through imitation learning, Q-Transformer adopts Q-learning methods. Alongside the TD error objective of Q-learning, a conservative regularizer is incorporated to ensure that the maximum value action remains in-distribution. This approach allows Q-Transformer to leverage not only successful demonstrations but also failed trajectories for learning.

RT-Trajectory [97] adopts trajectory sketches as policy conditions instead of relying on language conditions or goal conditions. These trajectory sketches consist of curves that delineate the intended trajectory for the robot end-effector to follow. They can be manually specified through a graphical user interface, extracted from human demonstration videos, or generated by foundation models. RT-Trajectory's policy is built upon the backbone of RT-1 and trained to control the robot arm to accurately follow the trajectory sketches. This approach facilitates generalization to novel objects, tasks, and skills, as trajectories from various tasks are transferable.

ACT [98] builds a conditional VAE policy with action chunking, requiring the policy to predict a sequence of actions rather than a single one. During inference, the action sequence is averaged using a method called temporal ensembling. RoboAgent [99] extends ACT via MT-ACT, demonstrating that action chunking improves temporal consistency. It also introduces an inpainting-based semantic augmentation method.

RoboFlamingo [100] adapts the OpenFlamingo VLM to a robot policy by attaching an LSTM-based policy head. This demonstrates that pretrained VLMs can be effectively transferred to language-conditioned robotic manipulation tasks.

3) Control Policies for Multimodal Instructions: Multimodal instruction enables new ways to specify tasks, such as through demonstrations, by naming novel objects, or by pointing with a finger or mouse click.

VIMA [125] places a significant emphasis on multimodal prompts and the generalization capabilities of models. By incorporating multimodal prompts, more specific and intricate tasks can be formulated compared with pure text prompts. VIMA introduces six main types of tasks: object manipulation, visual goal reaching, novel concept grounding, one-shot video imitation, visual constraint satisfaction, and visual reasoning. These tasks are often challenging or even infeasible to express using only language prompts. VIMA-Bench has been developed to evaluate across four generalizability levels: placement, novel combinatorial, novel object, and novel task.

MOO [94] extends RT-1 to handle multimodal prompts. Leveraging the backbone of RT-1, it incorporates OWL-ViT to localize target objects specified by the prompt. By expanding the RT-1 dataset with new objects and additional prompt images, MOO enhances the generalization capabilities of RT-1. This extension also facilitates new methods of specifying target objects, such as pointing with a finger and clicking on graphical user interfaces.

4) Control Policies With 3-D Vision: In our 3-D world, it is reasonable to assume that 3-D vision provides richer information than 2-D images.

Point clouds are a common representation for 3-D visual inputs due to their straightforward derivation from RGB-D inputs, as demonstrated by both DP3 [106] and 3-D Diffuser Actor [109]. Voxels have likewise been extensively studied. VER [126] proposes voxelizing multiview images into 3-D cells in a coarse-to-fine manner, enhancing performance in vision–language navigation tasks. PerAct [88] facilitates efficient task learning with only a few demonstrations by leveraging voxel representations in both the observation and action spaces. The input comprises voxel maps reconstructed from multiview RGB-D images, while the output corresponds to the best voxel for guiding the gripper's movement. RoboUniView [101] improves performance by injecting 3-D information from multiperspective images through a novel UVFormer vision encoder, which is pretrained on a 3-D occupancy task.

In contrast, Act3D [89] introduces a continuous resolution 3-D feature field, with adaptive resolutions based on the current task, addressing the computational cost of voxelization. RVT, RVT-2 [90], [91] propose re-rendering images from virtual views of the scene point cloud and using these images as inputs, rather than directly relying on 3-D inputs.

5) Diffusion-Based Control Policies: Diffusion-based action generation leverages the success of diffusion models in the field of CV.

Diffusion Policy [105] formulates a robot policy as a DDPM. This approach incorporates a variety of techniques, including receding horizon control, visual conditioning, and the time-series diffusion Transformer. The effectiveness of this diffusion-based visuomotor policy is underscored by its pro-

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ficiency in multimodal action distributions, high-dimensional action spaces, and training stability.

SUDD [107] presents a framework where an LLM guides data generation, and subsequently, the filtered dataset is distilled into a visuo-linguo-motor policy. This framework achieves language-guided data generation by composing an LLM with a suite of primitive robot utilities, such as grasp samplers and motion planners. It then extends Diffusion Policy by incorporating language-based conditioning for multitask learning and facilitates the distillation of the filtered dataset.

Octo [108] introduces a Transformer-based diffusion policy characterized by a modular open-framework design, allowing for the flexible integration of different task definition encoders, observation encoders, and action decoders with the Octo Transformer. Being among the first to utilize the OXE dataset [113], Octo demonstrates positive transfer and generalizability across diverse robots and tasks.

MDT [110] builds a variant of the DiT model from CV for the action prediction head. Originally proposed as a Transformer-based diffusion model, DiT replaced the classical U-Net architecture in image generation. Coupled with two auxiliary objectives—masked generative foresight and contrastive latent alignment—MDT demonstrates better performance than prior U-Net-based diffusion models.

RDT-1B [111] is a diffusion-based foundation model for bimanual manipulation, also built on DiT. It addresses data scarcity by introducing a physically interpretable unified action format across various robots, enabling pretraining on heterogeneous multirobot datasets with over 1M trajectories. This allows RDT to scale up to 1.2B parameters and demonstrate zero-shot generalization.

6) Diffusion-Based Control Policies With 3-D Vision: Several works have proposed combining 3-D vision with diffusion-based policies. DP3 [106] introduces 3-D point cloud inputs to a diffusion policy. Similarly, 3-D diffuser actor [109] shares the core idea of DP3 but differs in model architecture by combining Act3D with Diffusion Policy. 3D-MoE [127] explores an efficient mixture-of-experts architecture with DiT-based action diffusion using a rectified flow framework.

7) Control Policies for Motion Planning: Motion planning involves decomposing movement tasks into discrete waypoints while satisfying constraints such as obstacle avoidance and kinematic limits.

The *language costs* framework [85] presents a novel approach to robot correction using natural language for human-in-the-loop control. This method leverages predicted cost maps generated from human instructions, which are then utilized by the motion planner to compute the optimal path. This framework enables users to correct goals, specify preferences, or recover from errors through intuitive language commands.

VoxPoser [104] employs an LLM and a VLM to create 3-D voxel maps that represent affordances and constraints. It leverages the programming capability of LLMs and the perception capability of VLMs. The LLM translates language instructions into executable code, invoking the VLM to obtain object coordinates. Based on the composed affordance and constraint maps, VoxPoser employs model predictive control to generate a feasible trajectory for the robot arm's end-effector. Notably, VoxPoser does not require any training, as it directly connects LLMs and VLMs for motion planning.

RoboTAP [128] breaks down demonstrations into stages marked by the opening and closing of the gripper. In each stage, RoboTAP uses the TAPIR algorithm to detect active points that track the relevant object from the source to the target pose. These tracked points are then used by a visual servoing controller to guide the robot. A motion plan is created by chaining these stages together, enabling few-shot visual imitation.

8) Control Policies With Point-Based Actions: Recent research has explored leveraging the capabilities of VLMs to select or predict point-based actions—a cost-effective alternative to building full VLAs.

PIVOT [129] casts robotics tasks as a VQA problem, leveraging VLMs to select the best robot action from a set of visual proposals. Visual proposals are annotated in the form of keypoints on images. The VLM is iteratively prompted to refine them until the best option is identified.

RoboPoint [92] finetunes a VLM on the task of spatial affordance prediction, in which the model outputs 2-D points indicating where to act on the image. These affordance points are subsequently projected into 3-D space using depth maps, forming the predicted robot action.

ReKep [36] introduces constraint functions that map 3-D keypoints in the scene to a numerical cost. Robotics manipulation tasks can be represented as a sequence of these constraints, which are generated by large vision models and VLMs. Consequently, robot actions can be obtained by solving the resulting constrained optimization problem.

9) Large VLA: LVLA is equivalent to the original VLA definition proposed by RT-2 [1], as illustrated in Fig. 2(a). This terminology is analogous to the distinction between LLMs and general language models, or between large VLMs and general VLMs.

RT-2 [1] leverages the capabilities of large multimodal models for robotics tasks, building upon architectures like PaLI-X and PaLM-E. The approach introduces a co-fine-tuning strategy, jointly training the model on both Internet-scale VQA data and robot data. This training scheme enhances the model's generalizability and enables emergent capabilities.

RT-H [112] introduces an action hierarchy that includes an intermediate prediction layer of language motions, situated between language instructions and low-level actions (translation and rotation). This additional layer facilitates improved data sharing across different tasks. For example, both the language instructions "pick" and "pour" may involve the language motion "move the arm up." Moreover, this action hierarchy enables users to specify corrections to recover from failures, from which the model can subsequently learn.

RT-X [113] builds upon the previous RT-1 and RT-2 models. These models are retrained using a newly introduced open-source dataset named open X-embodiment (OXE), which is orders of magnitude larger than previous datasets. The resulting models, RT-1-X and RT-2-X, both outperform their original versions.

OpenVLA [35] was later developed as an open-source counterpart to RT-2-X. It additionally explored efficient finetuning methods, including LoRA and model quantization.

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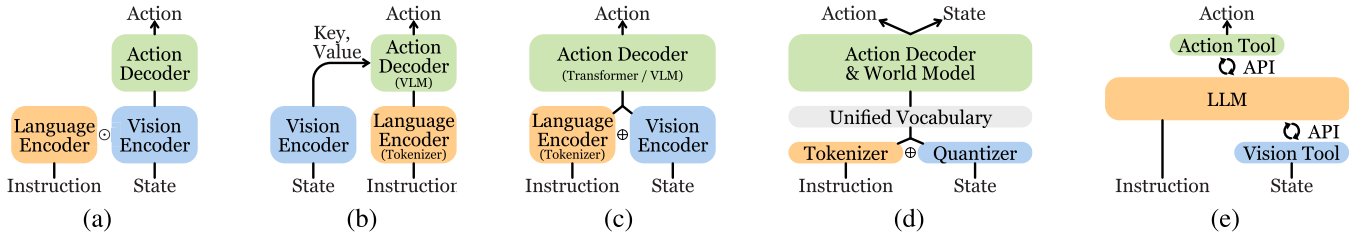

Fig. 5. Representative architectures of VLA models. ⊙: Hadamard product. ⊕: Concatenation. (a) FiLM. (b) Cross-attention. (c) Concatenation. (d) Quantization. (e) Tool use.

OpenVLA-OFT [114] applies an optimized fine-tuning (OFT) recipe for improved efficiency and performance. TraceVLA [115] finetunes OpenVLA to enable visual trace prompting, enhancing the model's spatial-temporal awareness.

π_0 [116] proposes a flow-matching architecture for transforming VLMs into VLAs. By incorporating an additional action expert based on the mixture-of-experts framework, it effectively inherits the Internet-scale knowledge of the base VLM while extending its capabilities to address robotic tasks.

RoboMamba [117] replaces the computationally expensive Transformer architecture with a Mamba state space model featuring linear inference complexity, thereby achieving efficient robotic reasoning and action capabilities.

SpatialVLA [118] introduces Ego3D position encoding for injecting 3-D information and represents actions with adaptive action grids for enhanced generalizability and robustness.

LAPA [122] devises the first unsupervised pretraining method for VLAs based on latent actions [69]. This approach employs a three-stage process to learn from Internet-scale unlabeled videos. First, a VQ-VAE framework is pretrained to extract quantized latent actions between image frames. Next, a VLA model is pretrained to predict these latent actions. Finally, the model is finetuned using only a small robotic dataset to map the latent actions to actual robot actions.

The integration of LVLAs with diffusion has also been gaining popularity. TinyVLA [119] leverages Diffusion Policy, while CogACT [120] utilizes a DiT action diffusion module. DexVLA [18] proposes embodied curriculum learning to progressively train a diffusion action expert, incorporating substep reasoning for decomposing long-horizon tasks. HybridVLA [121] integrates diffusion with the autoregressive paradigm to fully leverage VLMs' reasoning capabilities.

Visual autoregressive (VAR) modeling [130] with quantized visual tokens demonstrates improved performance over diffusion models in image generation. This suggests that the three modalities of VLAs can be unified under the autoregressive paradigm [131]. WorldVLA [123] and UniVLA [124] advance this direction by integrating VLAs with world models. They quantize multimodal data into discrete tokens, forming a shared vocabulary of quantized multimodal tokens. Consequently, all modalities can be modeled autoregressively, enabling not only action and text generation but also image generation, thereby constituting a world model.

A recent trend in LLMs is equipping them with tool-use capabilities by generating code that calls tools via APIs [132]. Instruct2Act [103] follows this paradigm by integrating vision and action tools, enabling LLMs to perform robotic tasks.

10) Strengths and Limitations:

a) Architectures: Representative VLA architectures are illustrated in Fig. 5. FiLM is used in RT-1, and thus its follow-up models inherit this mechanism. While cross-attention may offer superior performance with smaller model sizes, concatenation is simpler to implement and can achieve comparable results with larger models [125]. Quantization unifies multimodal tokens into a shared vocabulary, thus enabling integration with world models. The tool-use paradigm of LLMs can also be applied to robotic tasks.

b) Action Types and Their Training Objectives: Most low-level control policies predict actions for the end-effector pose while abstracting away the motion planning module that produces more fine-grained motions. While this abstraction facilitates better generalization to different embodiments, it also imposes limitations on dexterity.

The BC objective is used in imitation learning, with different variants for different action types. The BC objective for continuous actions can be written as

$$\mathcal{L}_{\text{Cont}} = \sum_t \text{MSE}\left(a_t, \hat{a}_t\right), \tag{4}$$

where MSE(·) stands for mean squared error, a_t is the action annotation from expert demonstrations, and $\hat{a}_t$ is the predicted action.

Discrete actions are achieved by dividing the action value range into a fixed number of bins. Their BC objective is

$$\mathcal{L}_{\text{Disc}} = \sum_t \text{CE}\left(a_t, \hat{a}_t\right), \tag{5}$$

where CE(·) stands for cross-entropy loss.

The BC objective for SE(2) actions is

$$\mathcal{L}_{\text{SE(2)}} = \text{CE}\left(a_{\text{pick}}, \hat{a}_{\text{pick}}\right) + \text{CE}\left(a_{\text{place}}, \hat{a}_{\text{place}}\right). \tag{6}$$

The DDPM objective in diffusion-based control policies is

$$\mathcal{L}_{\text{DDPM}} = \text{MSE}\left(\varepsilon^k, \varepsilon_\theta(a_t^k, k)\right), \tag{7}$$

where a_t^k is the action a_t corrupted by random noise ε^k at diffusion iteration k, and ε_θ is the noise prediction network, i.e., the VLA model.

While discrete action has demonstrated superior performance in RT-1 [95], Octo [108] argues that it leads to early grasping issues. SE(2) actions require the model to predict only the pick pose and the place pose of the end-effector, which are sufficient for many tabletop manipulation tasks. However, more complex tasks—such as "pouring water into a cup"—may require additional DoFs, thereby necessitating SE(3) actions [36]. Although point-based actions can be

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TABLE IV
HIGH-LEVEL TASK PLANNERS

Model	Vision Model	Language Model / VLM	Low-level Control Policy	Architecture	Require Training	Environments, Embodiments, Tasks, and Skills
PaLM-E [10]	ViT, OSRT	PaLM	Interactive Language, RT-1	Concat	✓	**Sim**: TAMP; **Sim&Real-Mani**: Language-Table; **Real**: SayCan setup
EmbodiedGPT [133]	EVA ViT-G/14	LLaMA-7B	MLP	Xattn	✓	**Sim-Mani**: Franka Kitchen, Meta-World
LEO [134]	ConvNeXt, PointNet++	Vicuna-7B	MLP	Concat	✓	**Sim-Mani**: CLIPort task subset; **Sim-Navi**: Habitat-web
3D-LLM [42]	CLIP, 3D-CLR, ConceptFusion	Flamingo, BLIP-2	DD-PPO	Concat	✓	**Sim-Navi**: Habitat
ShapeLLM [135]	ReCon++	LLaMA	–	Concat	✓	3D MM-Vet Benchmark
SayCan [9]	–	PaLM, FLAN	BC-Z, MT-Opt	–	✗	**Real-Navi&Mani** (EDR): office kitchen
Translated ⟨LM⟩ [136]	–	GPT-3, Codex	–	–	✗	**Sim**: VirtualHome
$(SL)^3$ [137]	–	T5-small	seq2seq	–	✓	**Sim-Navi**: ALFRED
Inner Monologue [8]	MDETR, CLIP	InstructGPT, PaLM	CLIPort, heuristics, SayCan policies	Language	✗	**Sim&Real-Mani**: Tabletop Rearrangement; **Real-Navi&Mani**: Kitchen Mobile Mani.
LLM-Planner [138]	HLSM object detector	GPT-3	HLSM	Language	✓	**Sim-Navi**: ALFRED
LID [139]	CNN	GPT-2	–	Lang., Embed	✓	**Sim**: VirtualHome
SMs [140]	CLIP, ViLD	GPT-3, RoBERTa	CLIPort	Lang., Code	✗	**Sim-Mani**: pick, place
ProgPrompt [141]	ViLD	GPT-3	API	Code	✗	**Sim**: VirtualHome
ChatGPT for Robotics [142]	YOLOv8	ChatGPT	API	Code	✗	**Real-Navi**: drone flight; **Sim-Navi**: AirSim, Habitat
CaP [143]	ViLD, MDETR	GPT-3, Codex	API	Code	✗	**Sim-Mani** & **Sim-Navi**: pick, place, etc.
DEPS [144]	CLIP	ChatGPT	MC-Controller, Steve-1	Code	✗	**Sim**: Minecraft
ConceptGraphs [145]	SAM, CLIP, LLaVA	GPT-4	API	Code (JSON)	✗	**Sim**: AI2-THOR; **Real** (Spot Arm): pick-place

coarse-grained, they are more easily obtained from VLMs in a zero-shot manner.

c) RT Series: RT-1 [95] inspired a series of "Robotic Transformer" models. The Transformer backbone surpasses previous RNN backbones by harnessing the higher capacity of Transformers to absorb larger robot datasets. Preceding RT-1 was BC-Z, which solely utilized MLP layers for action prediction. Subsequent to RT-1, several works emerged to introduce new capabilities. MOO adapted RT-1 to accommodate multimodal prompts. RT-Trajectory enabled RT-1 to process trajectory sketches as prompts. Q-Transformer utilized Q-learning to train RT-1. RT-2, based on ViT and LLM, introduced a completely different architecture from RT-1. RT-X retrained RT-1 and RT-2 with a significantly larger dataset, resulting in improved performance. Based on RT-2, RT-H [112] introduced action hierarchies for better data sharing.

d) LVLA vs. Generalized VLA: While LVLAs can greatly enhance instruction-following abilities because they can better parse user intentions, concerns arise regarding their training cost and deployment speed. Slow inference speed, in particular, can significantly impact performance in dynamic environments, as changes in the environment may occur during inference. Therefore, several methods have been proposed to improve efficiency. TinyVLA [119] focuses on inference speed and data efficiency through a smaller VLM and diffusion head for robot action. DeeR-VLA [102] proposes only partially activating the model via early-exit dynamic inference.

e) Scaling Laws: As in LLMs, scaling laws have also been observed in robotics [146], [147], revealing the importance of model size, dataset size, and the diversity of environments and objects. Further research in this area can guide the development of VLAs with robust in-the-wild generalization capabilities.

IV. TASK PLANNERS

A high-level task planner π_ϕ aims to decompose a complex task ℓ into a sequence of subtasks $\hat{\mathbf{p}} = [p_1, p_2, \ldots, p_N]$ (i.e., a task plan), where each p_i serves as an instruction to low-level control policies π_θ:

$$\hat{\mathbf{p}} \sim \pi_\phi(\mathbf{p} \mid \ell, s_t). \tag{8}$$

This process is sometimes referred to as task or subgoal decomposition and is closely related to task and motion planning (TAMP) and embodied decision making. When equipped with task planners, VLAs can complete more complex, long-horizon tasks, as illustrated in Fig. 4. Details are summarized in Table IV. Ideally, task plans should also incorporate optimal scheduling for these subtasks.

A. Monolithic Task Planners

A single LLM or multimodal LLM (MLLM) can typically generate task plans by employing a tailored framework or through finetuning on embodied datasets. We refer to these as *monolithic* models.

1) End-to-End Task Planners: Similar to LVLAs, task planners can be implemented as end-to-end MLLMs, leveraging their Internet-scale knowledge for task planning.

PaLM-E [10] integrates ViT and PaLM to create an embodied MLLM capable of performing high-level embodied reasoning tasks. Based on perceived images and high-level language instructions, PaLM-E generates a text plan that serves as instructions for low-level robotic policies. In a mobile manipulation environment, the generated plan is mapped to executable low-level instructions using SayCan [9]. As the low-level policy executes actions, PaLM-E can also replan based on changes in the environment. With PaLM as its backbone, PaLM-E can handle standard VQA tasks along with additional embodied VQA tasks.

EmbodiedGPT [133] introduces the embodied-former, which outputs task-relevant instance-level features. This is achieved by incorporating vision encoder embeddings and embodied planning information provided by an LLM. These instance features serve to inform the low-level policy about the immediate next action to take.

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2) End-to-End Task Planners With 3-D Vision: Some task planners also explore the use of 3-D vision. Because the majority of current MLLMs deal with images as visual inputs, they require architectural changes to incorporate 3-D visual inputs. Consequently, they are usually end-to-end models.

LEO [134] uses a two-stage training strategy to integrate a point cloud encoder with an LLM: the first stage focuses on 3-D vision–language alignment, followed by the second stage of 3-D vision-language-action instruction tuning. LEO performs well not only in 3-D question-answering tasks but also in manipulation, navigation, and task planning.

3D-LLM [42] injects 3-D information into LLMs and empowers them to perform 3-D tasks, such as 3-D-assisted dialog and navigation, using features from point clouds, gradSLAM, and neural voxel fields. MultiPLY [148] is an object-centric embodied LLM that incorporates even more modalities, including audio, tactile, and thermal.

ShapeLLM [135] is built on the novel 3-D vision encoder ReCon++, which distills knowledge from multiview image and text teacher models, along with a point cloud MAE. By integrating ReCon++ with LLaMA, ShapeLLM improves embodied interaction performance on their newly proposed 3-D benchmark, 3-D MM-Vet.

3) Grounded Task Planners: Grounded task planning involves generating high-level actions while considering whether they can be executed by low-level control policies.

SayCan [9] is a framework designed to integrate high-level LLM planners with low-level control policies. In this framework, the LLM planner accepts a user's high-level instruction and "says" what the most probable next low-level skill is, a concept referred to as *task-grounding*. The low-level policy provides the value function as the affordance function, determining the probability that the policy "can" complete the skill, known as *world-grounding*. By considering both the LLM's plan and the affordances, the framework selects the optimal skill for the current state.

Translated ⟨LM⟩ [136] employs a two-step process to translate high-level instructions into executable actions. Initially, a pretrained causal LLM is utilized for *plan generation*, breaking down the high-level instruction into the next action, expressed as a free-form language phrase. Then, as this phrase may not directly map to VirtualHome actions, a pretrained masked LLM is employed for *action translation*. This step involves calculating the similarity between the generated action phrase and the admissible VirtualHome actions. The translated action is appended to the plan, and the updated plan is fed back into the LLM to generate the next action phrase. This two-step process is repeated until a complete plan is formed. A "re-prompting" strategy [149] is further proposed to generate corrective actions when the agent encounters precondition errors.

$(SL)^3$ [137] is a learning algorithm that alternates between three steps: segmentation, labeling, and parameter update. In the segmentation step, high-level subtasks are aligned with low-level actions; subtask descriptions are then inferred in the labeling step; and finally, the network parameters are updated. This approach enables a hierarchical policy to discover reusable skills with sparse natural language annotations.

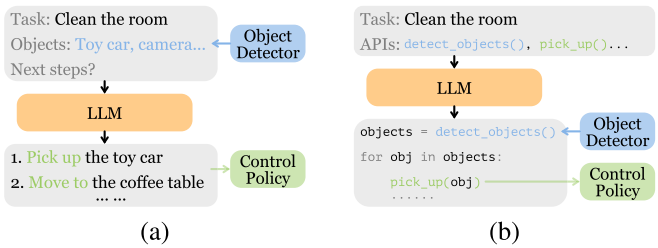

Fig. 6. Different approaches to connect LLM to multimodal modules in modular task planners. (a) Language-based. (b) Code-based.

B. Modular Task Planners

Finetuning end-to-end models on embodied data can be expensive; therefore, some approaches adopt a modular design by assembling off-the-shelf LLMs and VLMs into task planners. These approaches can also be viewed as following the tool-use architecture [132].

1) Language-Based Task Planners: Language-based approaches use natural language descriptions as the medium for exchanging multimodal information, as shown in Fig. 6(a).

Inner monologue [8] sits between high-level commands and low-level policies to enable closed-loop control planning. It employs an LLM to generate language instructions for low-level policies and dynamically updates these instructions based on the received feedback. The feedback encompasses various sources: success feedback, object and scene feedback, and human feedback. As the feedback is communicated to the LLM in a textual format, no additional training is required for the LLM. A similar approach is used in ReAct [76].

LLM-Planner [138] introduces a novel approach to constructing a hierarchical policy comprising a high-level planner and a low-level planner. The high-level planner harnesses the capabilities of an LLM to generate natural language plans, while the low-level planner translates each subgoal within the plan into primitive actions. While sharing similarities with previous methods in its overall architecture, LLM-Planner distinguishes itself by incorporating a re-planning mechanism, enabling the robot to "get unstuck."

LID [139] introduces a novel data collection procedure termed active data gathering (ADG). A key aspect of ADG is hindsight relabeling, which reassigns labels to unsuccessful trajectories, effectively maximizing the utilization of data irrespective of trajectory success. By converting all environmental inputs into textual descriptions, the proposed language-model-based policy gains enhanced combinatorial generalization.

Socratic models (SMs) [140] present a unique framework wherein diverse pretrained models are effectively composed without the need for finetuning. The framework relies on a key mechanism called multimodal-informed prompting, which facilitates information exchange among models with varied multimodal capabilities. The idea is to utilize multimodal models to convert non-language inputs into language descriptions, effectively unifying different modalities within the language space. Beyond excelling in conventional multimodal tasks, SMs showcase their versatility in robot perception and planning. In addition to natural language plans, task plans can also be represented in the form of pseudocode.

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2) Code-Based Task Planners: Code-based task planners leverage the coding ability of LLMs to generate task plans in the form of executable programs. Object detectors, VLMs, and control policies can be invoked via APIs, as shown in Fig. 6(b).

ProgPrompt [141] introduces a novel task-planning approach by prompting LLMs with program-like specifications detailing available actions and objects. This enables LLMs to generate high-level plans for household tasks in a few-shot manner. Environmental feedback can be incorporated through assertions within the program.

ChatGPT for robotics [142] takes advantage of the programming ability of ChatGPT to facilitate "user on the loop" control, a departure from the conventional "engineer in the loop" methodology. The procedure includes several steps. First, a list of APIs is defined, such as an object-detection API, a grasp API, and a move API. Second, a prompt is constructed for ChatGPT, specifying the environment, API functionality, and task goal. Third, ChatGPT is iteratively prompted to write code utilizing the defined APIs to execute the task, provided access to simulation and user feedback for evaluating code quality and safety. Finally, the ChatGPT-generated code is executed. In this procedure, ChatGPT serves as a high-level task planner, and actions are generated through function calls to corresponding low-level APIs.

Code as policies (CaP) [143] also leverages the code-writing capability of LLMs. It employs GPT-3 or Codex to generate policy code, which, in turn, invokes perception modules and control APIs. CaP exhibits proficiency in spatial-geometric reasoning, generalization to new instructions, and parameterization for low-level control primitives. By leveraging the multimodal capabilities of GPT-4V, COME-robot [150] eliminates the need for perception APIs required by CaP. This opens up possibilities for open-ended reasoning and adaptive planning within a closed-loop framework, enabling capabilities such as failure recovery and free-form instruction following.

DEPS [144] stands for "describe, explain, plan, and select." This approach employs an LLM to generate plans and explain failures based on feedback descriptions collected from the environment—a process referred to as "self-explanation," which aids in replanning. In addition, DEPS introduces a trainable goal selector to choose among parallel candidate subgoals based on how easily they can be achieved, a crucial aspect often overlooked by other high-level task planners.

ConceptGraphs [145] introduces a method to convert observation sequences into open-vocabulary 3-D scene graphs. Objects are extracted from RGB images using 2-D segmentation models, and VLMs are employed to caption objects and establish inter-object relations, resulting in the construction of a 3-D scene graph. This graph can then be translated into a text description (JSON), providing LLMs with rich semantic and spatial relationships between entities for task planning.

C. Strengths and Limitations

Monolithic task planners that utilize grounded task planning focus on generating executable plans. End-to-end models share an architecture similar to most LVLAs and can be finetuned on specialized embodied data to achieve better performance.

TABLE V

EMBODIED DATASETS. PER RT-X, SKILLS CORRESPOND TO VERBS, AND TASKS ARE DIFFERENT COMBINATIONS OF VERBS AND OBJECTS. * DATASET COLLECTED IN SIMULATION RATHER THAN THE REAL WORLD. SOME DATASETS ARE CONTINUALLY UPDATED, AND WE INCLUDE ONLY THE ORIGINAL VERSION. TABLE ADAPTED FROM [151] AND [152]

Dataset	Skills	Tasks	Scenes	Episodes	Collection	Obs.	Instruction	Robots
MIME [153]	12	20	1	8.3K	Human teleop.	RGBD	Demo	Baxter
*RoboTurk [154]	2	3	1	2.1K	Human teleop.	RGB	–	Sawyer
RoboNet [155]	–	–	10	162K	Script	RGB	Goal image	(7 robots)
MT-Opt [156]	2	12	1	800K	Script, RL	RGB	Task ID	(7 robots)
BC-Z [80]	3	100	1	25.9K	Human teleop.	RGB	Lang, demo	EDR
Fractal [95]	12	700+	2	130K	Human teleop.	RGB	Lang	EDR
MOO [94]	5	–	–	59.1K	Human teleop.	RGB	Multimodal	EDR
*VIMA [125]	17	–	1	650K	Script	RGB	Multimodal	UR5
RoboSet [157]	12	38	11	98.5K	Human, script	RGBD	Lang	Franka
BridgeV2 [158]	13	–	24	60.1K	Human, script	RGB(D)	Lang	WidowX
RH20T [151]	42	147	7	110K+	Human teleop.	RGBD	Lang	(4 robots)
DROID [152]	86	–	564	76K	Human teleop.	RGBD	Lang	Franka
OXE [113]	527	160K	311	1M+	Aggregate data	RGBD	Lang	(22 robots)

However, the training costs of such large models are often substantial. In contrast, *modular* task planners are more readily deployable because they leverage off-the-shelf LLMs and VLMs. Language-based task planners offer the advantage of seamless integration of LLMs and VLMs, as they are designed to operate in the natural language space. However, they often require extra steps to align the generated task plans with language instructions that are admissible to low-level control policies. Conversely, while code-based task planners may require manually wrapping VLMs and control policies within APIs and preparing clear documentation in advance, they enable code debugging and provide greater controllability. Ultimately, their performance remains constrained by the programming capabilities of the underlying models.

V. DATASETS AND BENCHMARKS

A. Real-World Robot Datasets and Benchmarks

Embodied AI faces significant data scarcity issues because real-world robot data are not as readily available as language data. Collecting real-world robot datasets poses multiple challenges. First, the process is impeded by the cost and time required to procure robotic equipment, set up environments, and gather expert data through dedicated policies or human teleoperation. Second, the diverse types and configurations of robots introduce inconsistencies in factors such as sensory data, control modes, and gripper types. Finally, accurately capturing 6-D object poses and reproducing setups remain elusive. We summarize recent embodied datasets in Table V. In addition, real-world benchmarks are further complicated by the need for human evaluation.

B. Simulators, Simulated Robot Datasets, and Benchmarks

Many researchers resort to simulated environments to circumvent real-world obstacles and scale the data collection process. We compare simulators and simulated benchmarks in Table VI. Nevertheless, this strategy presents its own challenges, chief among which is the sim-to-real gap. This discrepancy arises when models trained on simulated data exhibit poor performance during real-world deployment. The causes of this gap are multifaceted, encompassing unrealistic rendering quality, inaccuracies in physics simulations, and domain shifts characterized by object properties and robot motion planners. For instance, simulating nonrigid entities

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TABLE VI

SIMULATORS AND SIMULATED BENCHMARKS. CONTROL: CONTINUOUS CONTROL TASKS. *D*, *S*, *A*, AND *N*: DEPTH, SEGMENTATION, AUDIO, AND NORMAL, RESPECTIVELY. FORCE: SIMULATED CONTACT FORCE BETWEEN END-EFFECTOR AND ITEM. PD: PREDEFINED. TABLE ADAPTED FROM [159] AND [160]

Name	Scenes /Rooms	Objects /Cat	UI	Physics Engine	Task	Observation	Action	Agent	Description	Related
Gibson [161]	572/-	–	–	Pybullet	Navi	RGB, D, N, S	–	–	Navi only	–
iGibson [160], [162], [163]	15/108	152/5	Mouse, VR	Pybullet	Navi, Mani	RGB, D, S, N, Flow, LiDAR	Force	TurtleBot v2, LoCoBot, etc.	VR, Continuous Extended States. Versions: 0.5, 1.0, 2.0	Benchmarks: BEHAVIOR-100 [159], BEHAVIOR-1K [164]
SAPIEN [165]	–	2346/-	Code	PhysX	Navi, Mani	RGB, D, S	Force	Franka	Articulation, Ray Tracing	VoxPoser. Benchmark: SIMPLER [166]
AI2-THOR [167]	-/120	118/118	Mouse	Unity	Navi, Mani	RGB, D, S, A	Force, PD	ManipulaTHOR, LoCoBot, etc.	Object States, Task Planning. Versions: [168], [169]	Benchmarks: ALFRED, RoPOR [170]
VirtualHome [171]	7/-	-/509	Lang	Unity	Navi, Mani	RGB, D, S	Force, PD	Human	Object States, Task Planning	LID, Translated⟨LM⟩, ProgPrompt
TDW [172]	15/120	112/50	VR	Unity, Flex	Navi, Mani	RGB, D, S, A	Force	Fetch, Sawyer, Baxter	Audio, Fluids	–
RLBench [173]	1/-	28/28	Code	V-REP	Mani	RGB, D, S	Force	Franka	Tiered Task Difficulty	Hiveformer, PerAct
Meta-World [174]	1/-	80/7	Code	MuJoCo	Mani	Pose	Force	Sawyer	Meta-RL	R3M, VC-1, Vi-PRoM, EmbodiedGPT
CALVIN [175]	4/-	7/5	–	Pybullet	Mani	RGB, D	Force	Franka	Long-horizon Lang-cond tasks	GR-1, HULC, RoboFlamingo
Franka Kitchen [176]	1/-	10/6	VR	MuJoCo	Mani	Pose	Force	Franka	Extended by R3M with RGB	R3M, Voltron, Vi-PRoM, Diffusion Policy, EmbodiedGPT
Habitat [177], [178]	(Matterport + Gibson)		Mouse	Bullet	Navi	RGB, D, S, A	Force	Fetch, Franka, AlienGO	Fast, Navi only. Versions: 1.0, 2.0, Rearrangement [179]	VC-1, PACT; Benchmark: OVMM [180]
ALFRED [181]	-/120	84/84	–	Unity	Navi, Mani	RGB, D, S	PD	Human	Diverse long-horizon tasks	$(SL)^3$, LLM-Planner
DMC [182]	1/-	4/4	Code	MuJoCo	Control	RGB, D	Force	–	Continuous RL	VC-1, SMART
OpenAI Gym [183]	1/-	4/4	Code	MuJoCo	Control	RGB	Force	–	Single agent RL environments	–
Genesis [184]	(Rigid, deformable, liquid, etc.)		Code	(Custom)	Navi, Mani	RGB, D, S, N	Force	Franka, Unitree, etc.	High-speed comprehensive physics simulation	–

such as deformable objects or liquids presents significant difficulties. Moreover, importing new objects into simulators requires considerable effort, often involving techniques such as 3-D scanning and mesh editing. Despite these hurdles, simulated environments provide automated evaluation metrics that aid researchers in consistently evaluating robotic models. Many benchmarks are based on simulators because they can precisely reproduce the experimental setup and yield fair comparisons of different models. Another technique, known as real-to-sim, can improve simulation fidelity, recreate failure cases, or facilitate digital twins.

C. Automated Dataset Collection

Several approaches advocate for automated dataset collection. RoboGen [185] employs a generative simulation paradigm that proposes interesting skills, simulates corresponding environments, and selects optimal learning approaches to train policies for acquiring those skills. AutoRT [186] functions as a robot orchestrator driven by LLMs, generating tasks, filtering them by affordance, and utilizing either autonomous policies or human teleoperators to collect and evaluate data. DIAL [187] focuses on augmenting language instructions in existing datasets using VLMs. RoboPoint [92] generates scenes procedurally with randomized 3-D layouts, objects, and camera viewpoints.

D. Human Datasets

An alternative strategy to address data scarcity in real-world settings is to leverage human data. Human behavior offers plentiful guidance for robot policies due to its dexterity and diversity [188]. However, this strategy also comes with inherent limitations. Capturing and transferring human hand and body motions to robot embodiments is difficult. Moreover, the inconsistency in human data poses a hurdle, as some data are egocentric while others are captured from third-person perspectives. In addition, filtering human data to extract useful information can be labor-intensive. These obstacles underscore the complexities involved in incorporating human data into robot learning processes. UMI [189] proposes a method to mitigate these issues using hand-held grippers. For a more comprehensive comparison of human datasets, we refer interested readers to [190].

E. Task Planning Benchmarks

EgoPlan-Bench [191] focuses on benchmarking real-world task planning with human annotations. PlanBench [192], [193] comprehensively assesses various aspects of task planning ability, such as cost optimality, plan verification, and replanning. LoTa-Bench [194] directly evaluates task planning by executing the generated plans in simulators and calculating success rates. Embodied agent interface (EAI) [195] argues that this approach fails to pinpoint issues in LLMs. By formalizing the input-output of LLM-based modules for decision-making tasks, EAI enables more fine-grained metrics beyond success rates.

F. Embodied Question Answering (EQA) Benchmarks

EQA benchmarks, as summarized in Table VII, do not directly evaluate robotic tasks like manipulation and navigation, but instead target other relevant abilities for embodied AI, such as spatial reasoning, physics understanding, and world knowledge. EQA is akin to traditional VQA benchmarks, but differs in that the agent can actively explore the environment before providing an answer. EmbodiedQA [196] and IQUAD [197] were among the first works to introduce this type of benchmark. MT-EQA [198] focuses on complex questions involving multiple targets. MP3D-EQA [199] converts previous RGB inputs to point clouds, testing 3-D perception capabilities.

Active exploration requires access to a simulator, thereby restricting the ability to utilize real-world data. EgoVQA [200] shifts the focus of VQA to egocentric videos. EgoTaskQA [201] emphasizes spatial, temporal, and causal relationship reasoning. EQA-MX [202] investigates multimodal expressions (MXs), including regular verbal utterances and nonverbal gestures like eye gaze and pointing. OpenEQA [203] evaluates seven main categories, including functional reasoning and world knowledge.

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TABLE VII
EQA BENCHMARKS. EXPLORE: ACTIVE EXPLORATION

Benchmark	QA Pairs	Scenes	Source	Answer Type	Collection	Explore	Metrics
EQA [196]	5K	750 envs	House3D simulator	Answer set (172 answers)	Template	✓	Accuracy
IQUAD [197]	75K	30 rooms	AI2-THOR	Multiple choice	Template	✓	Accuracy
MT-EQA [198]	19.3K	588 envs	House3D simulator	Binary answer	Template	✓	Accuracy
MP3D-EQA [199]	1,136	83 envs	MatterPort3D	Answer set (53 answers)	Template	✓	Accuracy
EgoVQA [200]	600	16 videos	IU Multi-view	Multiple choice (1 out of 5)	Human annotators	✗	Accuracy
EgoTaskQA [201]	40K	2K videos	LEMMA dataset	Open answer, binary verification	Human annotators & template	✗	Accuracy
EQA-MX [202]	8.2M	750K images	CAESAR simulator	Answer set	Question templates, answer set	✗	Accuracy
OpenEQA [203]	557 + 1079	180 envs	HM3D + ScanNet	Open answer	Human annotators	✗	LLM Score

VI. CHALLENGES AND FUTURE DIRECTIONS

A. Safety First

Safety is paramount in robotics, as robots interact directly with the physical world. Ensuring the safety of robotic systems requires the integration of real-world commonsense. This involves the incorporation of robust safety guardrails, risk assessment frameworks, and human-robot interaction protocols. RLHF and "evaluation without execution" can also significantly lower safety risks [20]. Interpretability and explainability of VLA decision-making processes are also crucial for enhancing robot safety through error diagnosis and troubleshooting.

B. Datasets and Benchmarks

In addition to the issues discussed in Section V, comprehensive benchmarks that cover a wide range of skills, objects, embodiments, and environments remain to be developed. Moreover, metrics beyond the success rate are needed for a fine-grained diagnosis of issues in VLA models, as highlighted by EAI [195] for LLMs.

C. Foundation Models and Generalization

VLA foundation models or robotic foundation models (RFMs) for embodied AI remain an open research topic primarily due to the diversity in embodiments, environments, and tasks. Many have made significant progress [111], [116], but still lack generalization capability on par with LLMs in NLP. Attaining such a level of generalization is very challenging, as it requires the development of many core AGI capabilities.

D. Multimodality

VLAs inherit many challenges associated with multimodal models, such as obtaining useful embeddings and aligning different modalities. Current approaches, like ImageBind [204] and LanguageBind [205], align different modalities to the image or language embedding space, respectively. Within a unified embedding space, MLLMs can accommodate diverse modalities and become more general. However, whether focusing on embeddings alone is sufficient remains under debate. Although beyond the scope of VLAs, other modalities—such as audio [45], haptics [206], and gaze [207]—have proven useful for certain embodied AI applications. For instance, modeling human gaze data using an additional gaze network [208] or incorporating it via an auxiliary loss [209] has been shown to enhance the performance of the primary policy network. These approaches demonstrate that RL policies can benefit from human-like visual attention, as human gaze often reveals the most salient locations in the environment [210]. While incorporating additional modalities is often advantageous, it inevitably introduces extra complexity into the model design.

E. Framework for Long-Horizon Tasks

The hierarchical framework is currently the most practical approach for long-horizon tasks. However, it increases system complexity and potential points of failure. Frequent task execution failures can trigger replanning, which can cause significant latency. Moreover, monolithic task planners share similar architectures with LVLAs, so employing two large models can be redundant and may hinder scalability. In addition, although modular task planners typically do not require training, they are not plug-and-play: language-based models may generate subtasks that control policies cannot execute, whereas code-based models require modules to be prewrapped in APIs manually. Therefore, developing a unified framework that directly translates long-horizon tasks into low-level control signals in an end-to-end fashion is worth exploring.

F. Real-Time Responsiveness

Unlike conversational AI, many robotic applications require real-time decision-making to respond to dynamic environments. If inference time cannot keep pace with environmental changes, the model may generate obsolete actions repeatedly. However, current VLA models—especially LVLAs and task planners—face a tradeoff between speed and capacity. Novel mechanisms are thus needed to strike an optimal balance.

G. Multiagent Systems

Cooperative multiagent systems offer benefits such as distributed perception and collaborative fault recovery. However, they also face challenges, including effective communication, coordinated dispatching, and fleet heterogeneity. In certain scenarios, individual agents may have conflicting goals, which further increases complexity.

H. Ethical and Societal Implications

Robotics has always raised various ethical, societal, and legal concerns. These include risks related to privacy, job displacement, decision-making bias, and the impact on social norms and human relationships.

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I. Applications

Most current VLAs focus on household or industrial settings, but a wider range of applications is possible, such as virtual assistants, autonomous vehicles, and agricultural robots. Various embodiments may also call for specialized VLAs, including dexterous hands, drones, quadruped robots, and humanoid robots. One particularly important field is healthcare, encompassing surgical robots [211] and care robots [212]. Healthcare demands higher safety and privacy standards and may necessitate novel techniques such as human-in-the-loop (HITL) control and federated learning. Moreover, due to the significant domain gap, specialized vision models might be needed for medical images.

VII. Conclusion

VLA models hold immense promise for enabling embodied agents to interact with the physical world and fulfill user instructions. This article is the first survey to review LVLAs alongside generalized VLAs. Our taxonomy provides a high-level overview of three main lines of research: key components, control policies, and task planners. We meticulously analyze and compare their technical details, including model architectures, training strategies, and individual modules. In addition, we highlight essential resources for training and evaluating VLAs, such as datasets, simulators, and benchmarks. We hope this survey captures the rapidly evolving landscape of embodied AI and inspires future research.

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